IoT-BASED PLANT MONITORING SYSTEM

Using NodeMCU ESP8266, Environmental Sensors, Relay Module and Blynk

A PROJECT REPORT

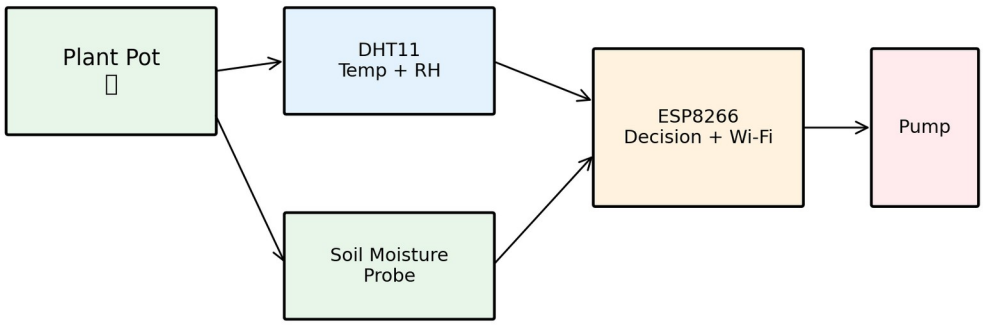
Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

in

Electronics and Communication Engineering

Figure 5: Sensor-to-Actuator Concept



Conceptual view of sensor-based plant monitoring

Submitted by: Ankit Gupta

Roll No.: _____

Under the Guidance of: _____

Department of Electronics and Communication Engineering

Academic Session: 2026–27

CERTIFICATE

This is to certify that the project report entitled **“IoT-Based Plant Monitoring System Using NodeMCU ESP8266 and Blynk”** is a bona fide work carried out by Ankit Gupta in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering.

The work presented in this report has been completed under the guidance and supervision of the undersigned and is suitable for academic evaluation.

Guide Signature: _____
Name: _____
Designation: _____

Head of Department: _____
Date: _____
Place: _____

DECLARATION

I hereby declare that the project report entitled “IoT-Based Plant Monitoring System Using NodeMCU ESP8266 and Blynk” is an original work carried out by me under the guidance of my project supervisor.

The project has been prepared for academic purposes and has not been submitted, in whole or in part, for the award of any other degree or qualification.

Date: _____
Place: _____

Signature of Student: _____
Name: Ankit Gupta

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I express my sincere gratitude to my project guide, faculty members and the Department of Electronics and Communication Engineering for their guidance, encouragement and technical support during the development of this project.

I am thankful to my institution for providing the laboratory facilities and resources required for implementing the IoT-based plant monitoring system. I also acknowledge the open-source Arduino ecosystem and Blynk IoT platform, which made it possible to develop and test the embedded monitoring and dashboard functions.

Finally, I thank my family and friends for their continuous encouragement and support.

Ankit Gupta

ABSTRACT

The IoT-Based Plant Monitoring System is designed to monitor important environmental parameters affecting plant growth and to automate irrigation when the soil becomes dry. The proposed system uses a NodeMCU ESP8266 microcontroller as the central controller. A DHT11 sensor measures temperature and relative humidity, while a soil-moisture sensor measures the moisture condition of the soil. The sensor values are processed by the NodeMCU and transmitted through Wi-Fi to the Blynk IoT platform.

A relay module is used as an interface between the low-voltage controller and a DC water pump. When the measured soil moisture falls below a selected threshold, the controller can activate the relay and switch the pump ON. When adequate moisture is restored, the pump is switched OFF. The Blynk dashboard provides remote monitoring and can also provide manual pump control.

The system demonstrates the integration of embedded systems, sensors, wireless communication, cloud-connected dashboards and automatic control. It can be adapted for home gardens, nurseries, greenhouses and small-scale smart agriculture.

Keywords: Internet of Things, NodeMCU ESP8266, DHT11, soil moisture, relay, water pump, Wi-Fi, Blynk, smart irrigation.

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1. INTRODUCTION

1.1 Background

Plant health depends strongly on temperature, humidity and soil-water availability. Traditional watering methods are often based on fixed schedules or manual observation. Such methods may result in overwatering, underwatering and unnecessary water consumption.

The Internet of Things (IoT) enables physical devices to collect data, communicate through networks and provide information to users remotely. By combining sensors, a Wi-Fi-enabled controller and a cloud dashboard, plant conditions can be monitored continuously.

1.2 Proposed System

The proposed system uses NodeMCU ESP8266 to read DHT11 temperature/humidity data and soil-moisture data. The ESP8266 connects to a Wi-Fi network and sends measurements to Blynk. A relay output controls a DC pump so that irrigation can be automated according to soil moisture.

1.3 Problem Statement

Manual plant monitoring requires regular physical inspection and may not respond quickly to changes in soil moisture. The project addresses this problem by providing remote sensing and automated watering.

2. OBJECTIVES

- Monitor plant temperature, relative humidity and soil moisture.
- Transmit sensor data to a Blynk IoT dashboard through Wi-Fi.
- Provide automatic ON/OFF control of a water pump using a relay.
- Provide an option for manual pump control from the dashboard.
- Develop the complete system using Arduino IDE and NodeMCU ESP8266.
- Create a low-cost and expandable prototype for smart gardening.

2.1 Scope

The prototype is intended for a single plant or small planting area. The design can be expanded by adding more moisture sensors, additional relays, larger pumps, weather sensors and multiple dashboard widgets.

3. INTERNET OF THINGS AND SYSTEM ARCHITECTURE

3.1 Internet of Things

IoT refers to connected physical devices that sense information, process it and exchange data over a communication network. In this project, the NodeMCU acts as the connected edge device. Sensors provide physical measurements, the ESP8266 performs local processing, Wi-Fi provides network access and Blynk provides remote visualization and control.

3.2 System Architecture

Figure 1: IoT-Based Plant Monitoring System Architecture

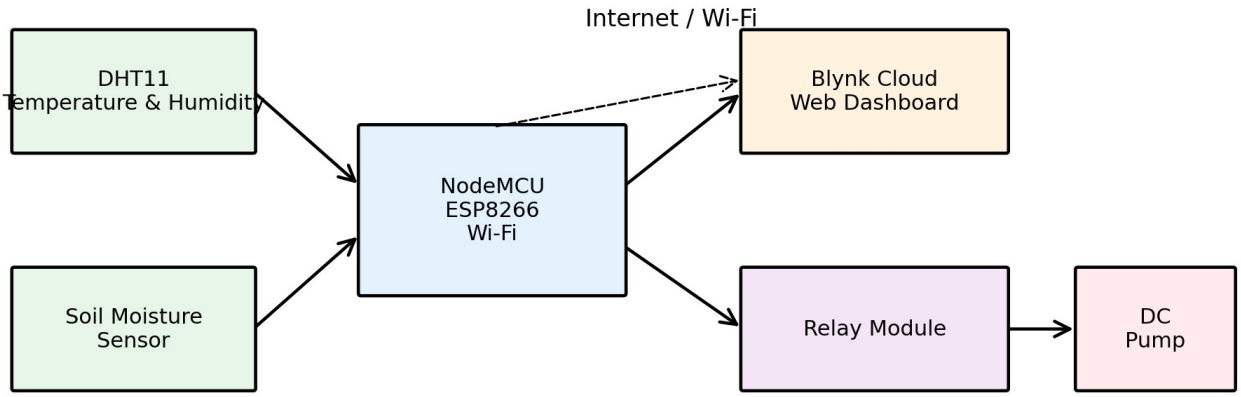


Figure 1: IoT-Based Plant Monitoring System Architecture

The system has five major functional sections: sensing, processing, communication, cloud/dashboard and actuation. The sensing section includes DHT11 and soil moisture sensing. NodeMCU reads and processes these values. Wi-Fi connects the controller to Blynk. The relay and pump form the actuation section.

4. HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware Requirements

S.No.	Component	Typical Specification	Purpose
1	NodeMCU ESP8266	ESP-12E/ESP-12F Wi-Fi board	Main controller and Wi-Fi
2	DHT11	Temperature + humidity sensor	Environmental measurement
3	Soil moisture sensor	Analog probe/module	Soil water measurement
4	Relay module	1-channel, suitable coil voltage	Pump switching
5	DC water pump	5 V / 6 V / 12 V as selected	Irrigation
6	Power supply	Suitable regulated supply	Controller/pump power
7	Jumper wires	Male/Female	Connections
8	Breadboard/PCB	Prototype board	Assembly

4.2 Software Requirements

- Arduino IDE
- ESP8266 board package for Arduino IDE
- Blynk IoT account and template/device
- Blynk library for ESP8266
- DHT sensor library and Adafruit Unified Sensor library

5. NODEMCU ESP8266

NodeMCU is an open-source development platform based on the ESP8266 Wi-Fi microcontroller. It is suitable for IoT prototypes because it combines a microcontroller, digital I/O, analog input and Wi-Fi communication on a compact development board.

5.1 Important Features

- Integrated 2.4 GHz Wi-Fi connectivity
- 32-bit Tensilica-based ESP8266 processor
- Digital GPIO pins for sensors and relay control
- Analog input available through A0
- Can be programmed using Arduino IDE
- Low-cost platform for IoT prototypes

5.2 Role in This Project

The NodeMCU reads the DHT11 and soil sensor, calculates a moisture percentage, sends data to Blynk and controls the relay. It therefore acts as both the local controller and the Internet-connected gateway.

6. SENSORS USED

6.1 DHT11 Temperature and Humidity Sensor

DHT11 is a digital sensor that provides temperature and relative humidity readings. In this project, the sensor is connected to a digital GPIO pin of the NodeMCU. A pull-up resistor may be required when using a bare DHT11 sensor; many module versions already include it.

6.2 Soil Moisture Sensor

The soil moisture sensor detects the moisture condition of soil. An analog module produces a voltage related to the probe condition, which is read through the NodeMCU A0 input. The raw value is mapped to a percentage after calibration.

Because sensor modules differ, the exact dry and wet ADC values should be measured experimentally. The software therefore contains calibration constants that can be adjusted for the actual sensor.

7. RELAY MODULE AND WATER PUMP

7.1 Relay Module

A relay is an electrically controlled switch. It allows a low-voltage microcontroller output to control a separate load circuit. The relay module used in the prototype receives a control signal from the NodeMCU and switches the pump supply.

7.2 Water Pump

A small DC pump is used to deliver water to the plant. The pump should be powered from an appropriate external supply. The NodeMCU must never be used as the direct power source for the pump because the pump current can exceed the controller's safe output capability.

7.3 Electrical Safety

- Use a relay/contact rating suitable for the pump voltage and current.
- Use an appropriate external power supply for the pump.
- Keep water away from exposed electrical connections.
- Use proper insulation and enclosure for practical deployment.
- Never connect a pump directly to an ESP8266 GPIO pin.

8. CIRCUIT DIAGRAM AND PIN CONNECTIONS

Figure 2: Circuit / Wiring Diagram

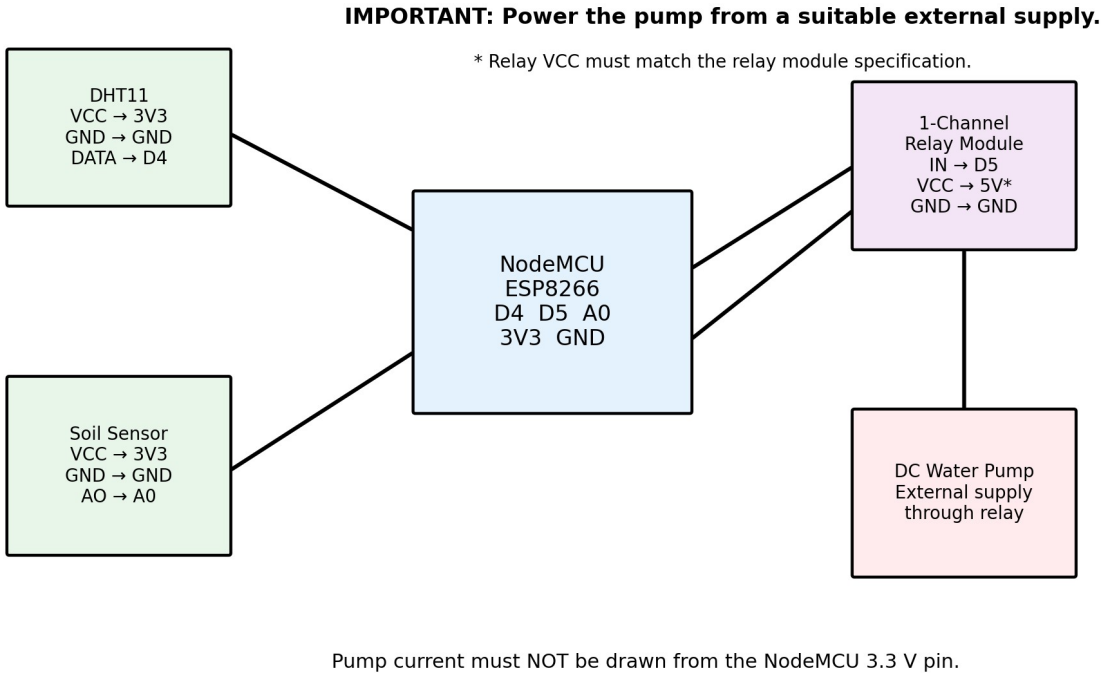


Figure 2: Circuit / Wiring Diagram

8.1 Suggested Pin Connections

Device	NodeMCU Pin	Connection
DHT11 DATA	D4 (GPIO2)	Digital input
Soil moisture AO	A0	Analog input
Relay IN	D5 (GPIO14)	Digital output
Sensor VCC	3V3	Sensor supply where supported
Sensor GND	GND	Common ground
Relay VCC	As specified by relay module	External/board supply as appropriate
Pump	Relay contacts	Separate pump supply

Note: ESP8266 ADC voltage limits depend on the development-board implementation. Verify the NodeMCU board's A0 input range and the soil-sensor output before connecting it.

9. BLYNK IoT WEB DASHBOARD

Blynk is used as the IoT dashboard for visualizing sensor data and providing remote control. The NodeMCU sends values to virtual datastreams. The dashboard can contain numeric displays, gauges, charts and a pump-control switch.

Figure 4: Example Blynk Web Dashboard Layout

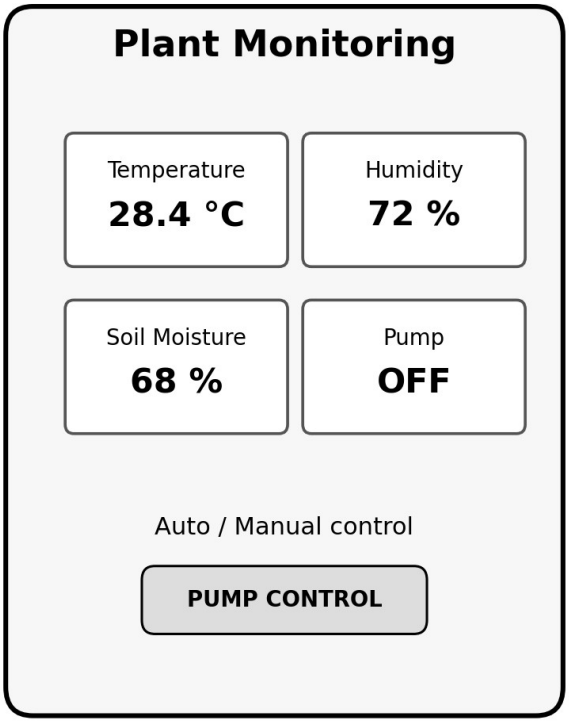


Figure 4: Example Blynk Web Dashboard Layout

9.1 Suggested Datastreams

Virtual Pin	Parameter	Example Value
V0	Temperature	28.4 °C
V1	Humidity	72 %
V2	Soil Moisture	68 %
V3	Pump Control	0/1

The exact widget names and dashboard layout can be changed according to the Blynk template used.

10. ARDUINO IDE AND PROGRAM CODE

The following program is a complete reference implementation. Replace the Blynk template credentials, Wi-Fi SSID and password with the values for the actual project.

```
// IoT Based Plant Monitoring System
// NodeMCU ESP8266 + DHT11 + Soil Moisture + Relay + Blynk

#define BLYNK_TEMPLATE_ID   "TMPLxxxxxxx"
#define BLYNK_TEMPLATE_NAME "Plant Monitor"
#define BLYNK_AUTH_TOKEN    "Your_Auth_Token"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>

char ssid[] = "Your_WiFi_Name";
char pass[] = "Your_WiFi_Password";

#define DHTPIN D4
#define DHTTYPE DHT11
#define SOIL_PIN A0
#define RELAY_PIN D5

// Calibrate these two values for the actual soil sensor.
int DRY_VALUE = 850;
int WET_VALUE = 350;

// Many relay modules are active LOW.
// Change to HIGH if your relay is active HIGH.
#define RELAY_ON  LOW
#define RELAY_OFF HIGH

DHT dht(DHTPIN, DHTTYPE);
BlynkTimer timer;

bool manualPump = false;
bool autoMode = true;

void sendSensorData()
{
  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();
  int rawSoil = analogRead(SOIL_PIN);

  if (isnan(temperature) || isnan(humidity)) {
    Serial.println("DHT11 read failed.");
    return;
  }

  int soilPercent = map(rawSoil, DRY_VALUE, WET_VALUE, 0, 100);
  soilPercent = constrain(soilPercent, 0, 100);

  Blynk.virtualWrite(V0, temperature);
  Blynk.virtualWrite(V1, humidity);
  Blynk.virtualWrite(V2, soilPercent);

  // Automatic irrigation
  if (autoMode) {
    if (soilPercent < 35) {
      digitalWrite(RELAY_PIN, RELAY_ON);
    }
    else if (soilPercent > 55) {
      digitalWrite(RELAY_PIN, RELAY_OFF);
    }
```

```
  }

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" C | Humidity: ");
  Serial.print(humidity);
  Serial.print(" % | Soil: ");
  Serial.print(soilPercent);
  Serial.println(" %");
}

// V3 = manual pump switch in Blynk
BLYNK_WRITE(V3)
{
  manualPump = param.asInt();

  if (!autoMode) {
    digitalWrite(RELAY_PIN, manualPump ? RELAY_ON : RELAY_OFF);
  }
}

// V4 = automatic/manual mode switch
BLYNK_WRITE(V4)
{
  autoMode = param.asInt();

  if (autoMode) {
    digitalWrite(RELAY_PIN, RELAY_OFF);
  } else {
    digitalWrite(RELAY_PIN, manualPump ? RELAY_ON : RELAY_OFF);
  }
}

void setup()
{
  Serial.begin(115200);

  pinMode(RELAY_PIN, OUTPUT);
  digitalWrite(RELAY_PIN, RELAY_OFF);

  dht.begin();

  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

  timer.setInterval(2000L, sendSensorData);
}

void loop()
{
  Blynk.run();
  timer.run();
}
```

Important: The Blynk Template ID, Template Name and Auth Token must match the actual Blynk device. Install the ESP8266 board package and required libraries before compiling.

11. WORKING PRINCIPLE

1. The DHT11 measures ambient temperature and relative humidity.

2. The soil-moisture sensor measures the condition of the soil and produces an analog signal.

3. NodeMCU reads both sensors and converts the soil sensor value into an approximate moisture percentage using calibration values.

4. The ESP8266 connects to the configured Wi-Fi network and communicates with the Blynk platform.

5. Sensor values are sent to the Blynk dashboard through virtual datastreams.

6. If automatic mode is enabled and soil moisture falls below the selected lower threshold, the relay is activated and the pump starts.

7. When soil moisture rises above the upper threshold, the pump is stopped. Using separate ON and OFF thresholds provides hysteresis and reduces rapid relay switching.

8. In manual mode, the user can operate the pump from the dashboard.

12. ALGORITHM AND FLOWCHART

Figure 3: System Flowchart

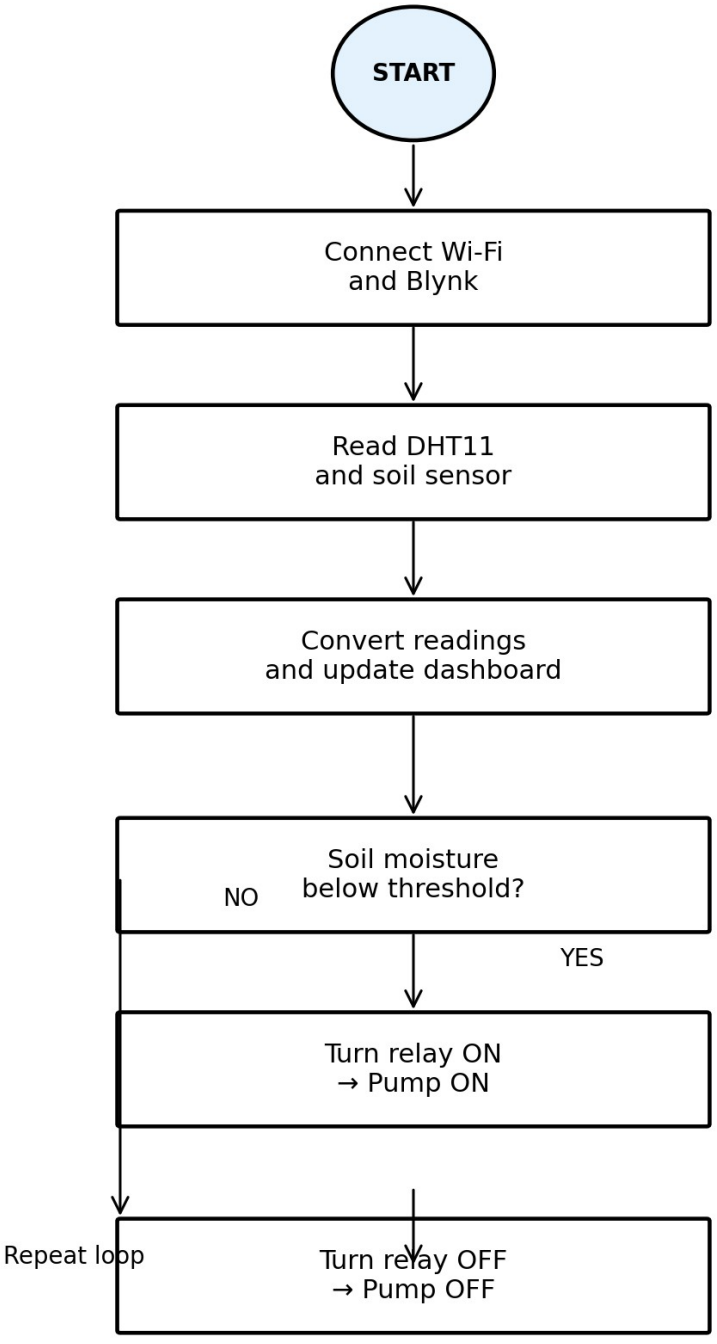


Figure 3: System Flowchart

12.1 Algorithm

1. Start the NodeMCU.

2. Initialize DHT11, soil sensor and relay pin.

3. Connect to Wi-Fi and Blynk.

4. Read temperature, humidity and soil moisture periodically.

5. Convert soil ADC reading to percentage using calibration constants.

6. Upload values to Blynk.

7. Check operating mode.

8. If automatic mode is selected, compare soil moisture with thresholds and control the relay.

9. If manual mode is selected, follow the pump-control command from Blynk.

10. Repeat continuously.

13. IMPLEMENTATION PROCEDURE

13.1 Hardware Assembly

Mount the NodeMCU, sensors and relay on a breadboard or prototype PCB. Connect the DHT11 data line to D4, the soil sensor analog output to A0 and the relay input to D5.

13.2 Sensor Calibration

Record the analog soil-sensor reading in dry soil and in well-watered soil. Enter these readings as DRY_VALUE and WET_VALUE in the program. Because sensors vary, calibration is necessary for meaningful percentage values.

13.3 Blynk Setup

Create a Blynk template/device, configure the required datastreams, add dashboard widgets and obtain the device authentication information.

13.4 Arduino IDE Setup

Install the ESP8266 board package, select the appropriate NodeMCU/ESP8266 board, install Blynk and DHT libraries, enter Wi-Fi/Blynk credentials and upload the program.

13.5 Testing

Open Serial Monitor, verify Wi-Fi connection, check sensor readings, observe dashboard updates and test the relay/pump first without water near electronics.

14. TESTING AND RESULTS

The following table is a sample observation format. Actual values should be filled from the physical prototype during demonstration.

Test	Temperature	Humidity	Soil Moisture	Pump	Result
1	28.4 °C	72 %	68 %	OFF	Normal
2	29.1 °C	69 %	42 %	OFF	Normal
3	30.0 °C	65 %	28 %	ON	Dry soil detected
4	29.4 °C	68 %	58 %	OFF	Moisture restored
5	27.8 °C	75 %	55 %	Manual	Manual test

14.1 Expected Result

The dashboard should show updated environmental values. In automatic mode, the pump should start when the moisture percentage crosses the lower threshold and stop after the upper threshold is reached. The relay should respond reliably without repeated rapid switching.

15. ADVANTAGES

- Remote monitoring through a Wi-Fi-connected dashboard.
- Automatic irrigation reduces dependence on manual watering.
- Combines multiple environmental parameters in one system.
- Low-cost and suitable for educational IoT prototypes.
- Thresholds can be changed in software.
- Expandable to additional sensors and actuators.
- NodeMCU provides built-in Wi-Fi, reducing the need for a separate communication module.

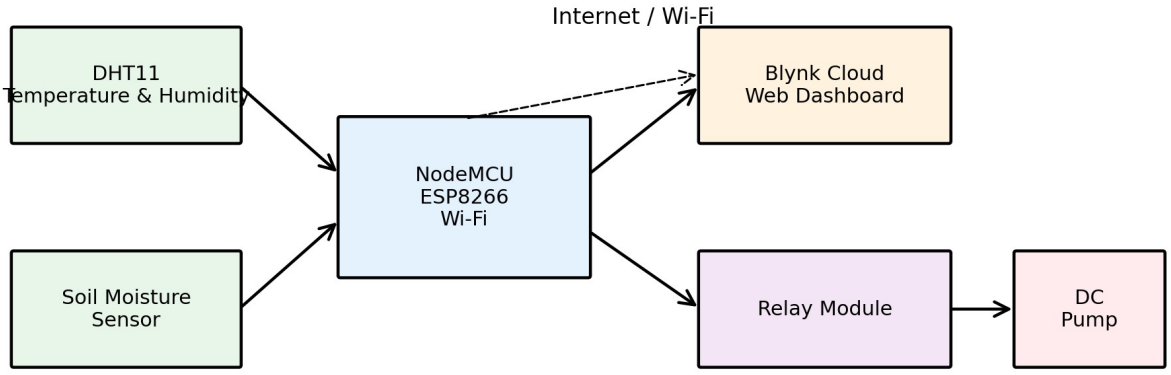
16. LIMITATIONS

- Wi-Fi and Internet availability are required for remote Blynk monitoring.
- Low-cost soil sensors may corrode or drift over time.
- Moisture percentage is an estimate unless the sensor is properly calibrated.
- Relay modules and pumps require suitable power supplies.
- DHT11 has lower accuracy and slower response than higher-grade environmental sensors such as DHT22 or industrial sensors.
- The prototype is intended for small-scale applications and needs additional protection for outdoor deployment.

17. APPLICATIONS

- Home gardens and indoor plants
- Greenhouses and nurseries
- Small-scale smart agriculture
- Plant growth experiments
- Educational IoT laboratories
- Remote environmental monitoring
- Automatic irrigation demonstration systems

Figure 1: IoT-Based Plant Monitoring System Architecture



System architecture applicable to smart gardening and small-scale agriculture

18. FUTURE SCOPE

- Replace DHT11 with DHT22/SHT31 for improved environmental sensing.
- Add light intensity and rain sensors.
- Use multiple soil-moisture sensors for several plants.
- Add a water-level sensor for the storage tank.
- Implement historical graphs and alerts for abnormal conditions.
- Add solar power and battery management for remote deployment.
- Use weather information to improve irrigation decisions.
- Design a waterproof enclosure and PCB for field installation.

19. CONCLUSION

The IoT-Based Plant Monitoring System demonstrates a practical integration of sensing, embedded processing, Wi-Fi communication, cloud-connected visualization and automatic actuation. NodeMCU ESP8266 provides the central control and communication functions, while DHT11 and soil-moisture sensing provide the required plant-environment information.

The relay-controlled pump enables automatic irrigation based on soil moisture. The Blynk dashboard makes the measurements available remotely and provides a convenient interface for manual control. The prototype therefore provides a useful foundation for smart gardening and can be expanded with additional sensors, better calibration, alerts and renewable power.

20. REFERENCES

1. ESP8266 / NodeMCU development-board documentation and technical references.

2. Arduino IDE documentation and ESP8266 Arduino core documentation.

3. DHT11 sensor datasheet and application information.

4. Soil moisture sensor module technical documentation.

5. Blynk IoT platform documentation and examples.

6. General references on Internet of Things (IoT), embedded systems and smart irrigation.

7. Manufacturer documentation for the selected relay module and DC water pump.

Note: Component specifications should be checked against the exact modules used in the physical prototype because different vendors may use different voltage levels, relay logic and sensor characteristics.

APPENDIX A: PROJECT COMPONENT CONNECTION SUMMARY

Component	Pin	NodeMCU	Purpose
DHT11	DATA	D4 / GPIO2	Temperature & humidity
Soil sensor	AO	A0	Soil moisture
Relay	IN	D5 / GPIO14	Pump switching
Blynk	V0	Virtual	Temperature
Blynk	V1	Virtual	Humidity
Blynk	V2	Virtual	Soil moisture
Blynk	V3	Virtual	Pump manual control
Blynk	V4	Virtual	Auto/manual mode

Recommended threshold example: Pump ON below 35% soil moisture and Pump OFF above 55%. These values are only starting points and must be calibrated for the actual soil, plant and sensor.

--- END OF PROJECT REPORT ---

PROTEUS CIRCUIT SIMULATION / SCHEMATIC

IoT-Based Plant Monitoring System

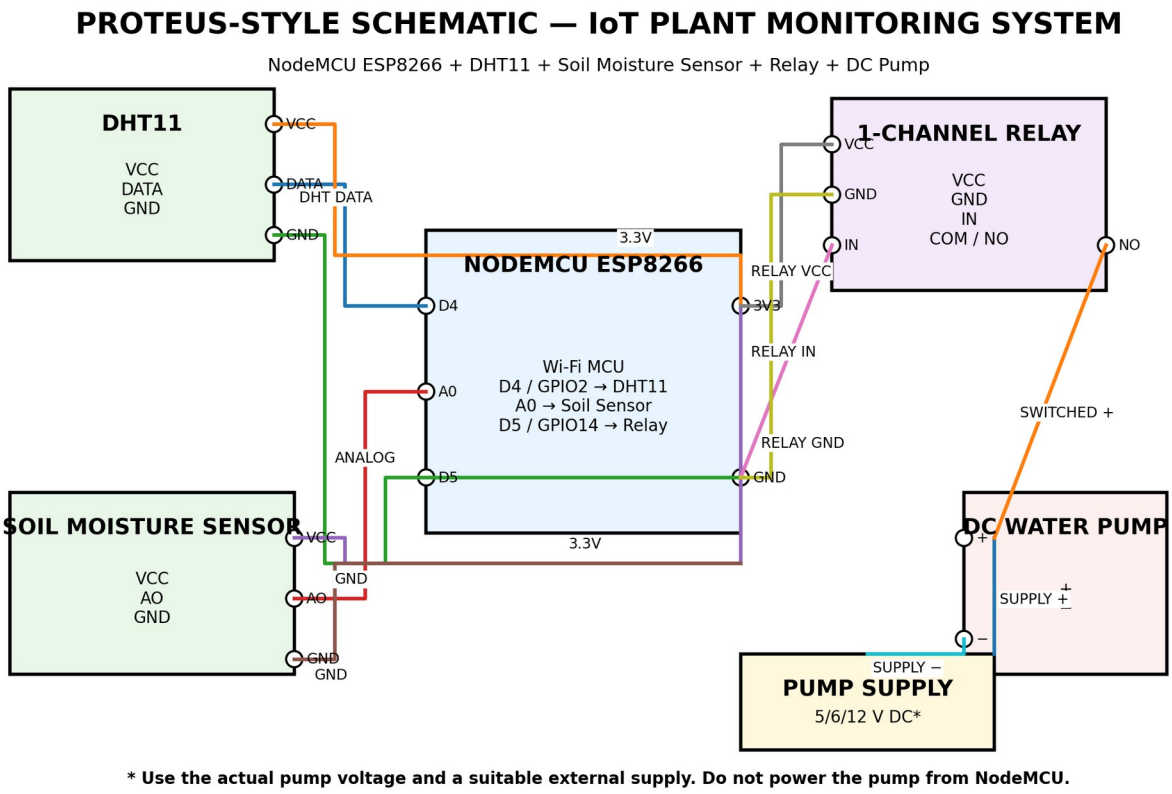


Figure 6: Proteus-style circuit schematic of the complete plant monitoring system

Circuit Connection Summary

Component	NodeMCU Pin	Connection	Function
DHT11 DATA	D4 / GPIO2	Digital input	Temperature & humidity
Soil Sensor AO	A0	Analog input	Soil moisture
Relay IN	D5 / GPIO14	Digital output	Pump switching
DHT11 / Sensor VCC	3V3	Supply	Sensor power
Sensors GND	GND	Common ground	Reference
Pump	Relay contacts	External supply	Irrigation

Note: This figure is a Proteus-style schematic representation prepared for the report. The exact Proteus library symbols and relay/pump models depend on the Proteus version and component libraries installed on the user's computer. Before physical implementation, verify the relay logic, NodeMCU A0 voltage range, sensor supply requirements and pump current rating.